applies basically to the atoms in a liquid, owing to the very similar properties of the forces responsible for their cohesion.

## IV. THE STABILITY OF NUCLEI

All that we have set forth above points to the important conclusion that the number 2—and in general, the even numbers—must have a preferred position in atomic nuclei. Consequently, we may expect those nuclei in which both the number of protons and the number of neutrons are even numbers to be particularly stable. For since every proton is able to bind two neutrons, and conversely, every neutron is able to bind two protons, in this case—and only in this case—all valencies can be utilized. This situation is, naturally, more advantageous from the point of view of energetics than one where the valency of an odd proton or neutron is not utilized.

But there is still another reason for the preferred position of the number 2: namely, Pauli's exclusion principle, which we discussed when dealing with the extranuclear structure of the atom. Stated in general terms, this principle asserts that in a stationary state (which with reference to the extranuclear atomic structure means a certain definite electronic orbit, or expressed in terms of the wave aspect, a certain constant vibration with a certain given direction of the electronic spin) there is room for one single particle only at any one time. Taking into consideration the fact that the spin of an electron may be either positive or negative—clockwise or counterclockwise—this can be expressed in the following form: No more than two electrons (with opposite spins) can occupy the same stationary orbit of the extranuclear structure of the atom. The same applies to the protons and neutrons in the nuclei, which likewise possess a spin momentum. It follows, therefore, that not more than two neutrons or two protons can occupy the same stationary orbit in a nucleus. From the point of view of energetics, it is of course more advantageous to have the possibility thus allowed fully utilized. This fact again results in a preferred position for nuclei containing an even number of neutrons and an even number of protons.

This preferred status of the number 2 is especially obvious in the common helium nucleus ${}_2He^4$, which consists of two neutrons and two protons. This is an especially stable structure, and so is its extranuclear electron structure, which consists of two electrons. This is demonstrated by the fact that helium is an inert gas and does not enter into any chemical combination whatsoever. Actually, the binding energy of the helium nucleus is extraordinarily high: roughly 30 Mev. On the other hand, the binding energy of the deuteron, composed of one neutron and one proton, is only 2·2 Mev., as already mentioned. In the deuteron, only one valency of the proton and one of the